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Design and Optimization of an Object Detection and Multi-Object Classification System using YOLOv8

Abstract

This report presents the design and implementation of an object detection system using the YOLOv8 (You Only Look Once) deep learning model.
The objective of this project is to analyze and implement a multi-object detection pipeline capable of identifying objects such as persons, vehicles,
and traffic signs in images. The model is trained using the COCO128 dataset and evaluated using standard metrics such as precision, recall,
and mean Average Precision (mAP). The results demonstrate the effectiveness of YOLOv8 for real-time object detection tasks while highlighting
the limitations associated with training on small datasets.

Introduction

Object detection is a fundamental task in computer vision where the goal is to identify objects in an image and locate them
using bounding boxes. With the rise of deep learning, object detection models such as YOLO have become extremely powerful and efficient.

YOLOv8, developed by Ultralytics, is one of the latest and most advanced versions of the YOLO family. It offers fast inference speed,
high accuracy, and easy deployment. In this project, YOLOv8 is used to build a multi-object detection pipeline capable of identifying
multiple objects in real-world scenes.

The project demonstrates the full workflow of object detection including dataset preparation, model training, evaluation, and inference.

Dataset Description

The dataset used in this experiment is COCO128, a small subset of the COCO dataset.
It contains 128 images annotated with bounding boxes for multiple object classes.

Classes used include:
Person, Bicycle, Car, Motorcycle, Airplane, Bus, Train, Truck, Boat, Traffic Light.

Although COCO128 is relatively small, it is suitable for demonstrating the YOLO training pipeline and evaluating the performance
of the object detection model.

Methodology

The methodology of the project follows the standard machine learning pipeline:

1. Dataset Collection – The COCO128 dataset was downloaded and organized.
2. Data Preparation – Images and annotations were structured in YOLO format.
3. Model Selection – YOLOv8n (nano model) was selected for faster training.
4. Training – The model was trained for multiple epochs using GPU acceleration.
5. Evaluation – Performance was evaluated using precision, recall, and mAP metrics.
6. Inference – The trained model was tested on new images to detect objects.

YOLOv8 processes images using a single neural network that predicts bounding boxes and class probabilities simultaneously.

Training Results

During training, multiple metrics were tracked including box loss, classification loss, and distribution focal loss.
The graphs show that losses gradually decrease over epochs while precision and recall improve.

This indicates that the model successfully learned to detect objects from the training data.

Evaluation Metrics

The model performance was evaluated using the following metrics:

Precision – Measures the accuracy of predictions.
Recall – Measures how many true objects were detected.
mAP (Mean Average Precision) – Evaluates overall detection performance.

For this experiment the model achieved approximately:

mAP50 ≈ 0.34
mAP50-95 ≈ 0.28

These values are reasonable considering the limited dataset size.

Conclusion

This project successfully demonstrated the implementation of an object detection system using YOLOv8.
The model was trained on the COCO128 dataset and evaluated using standard object detection metrics.

Although the dataset was small, the system was able to learn meaningful object representations and perform detection on test images.
Future work could include training on larger datasets, improving model accuracy, and deploying the system in real-time applications
such as traffic monitoring or surveillance systems.

Experimental Results:

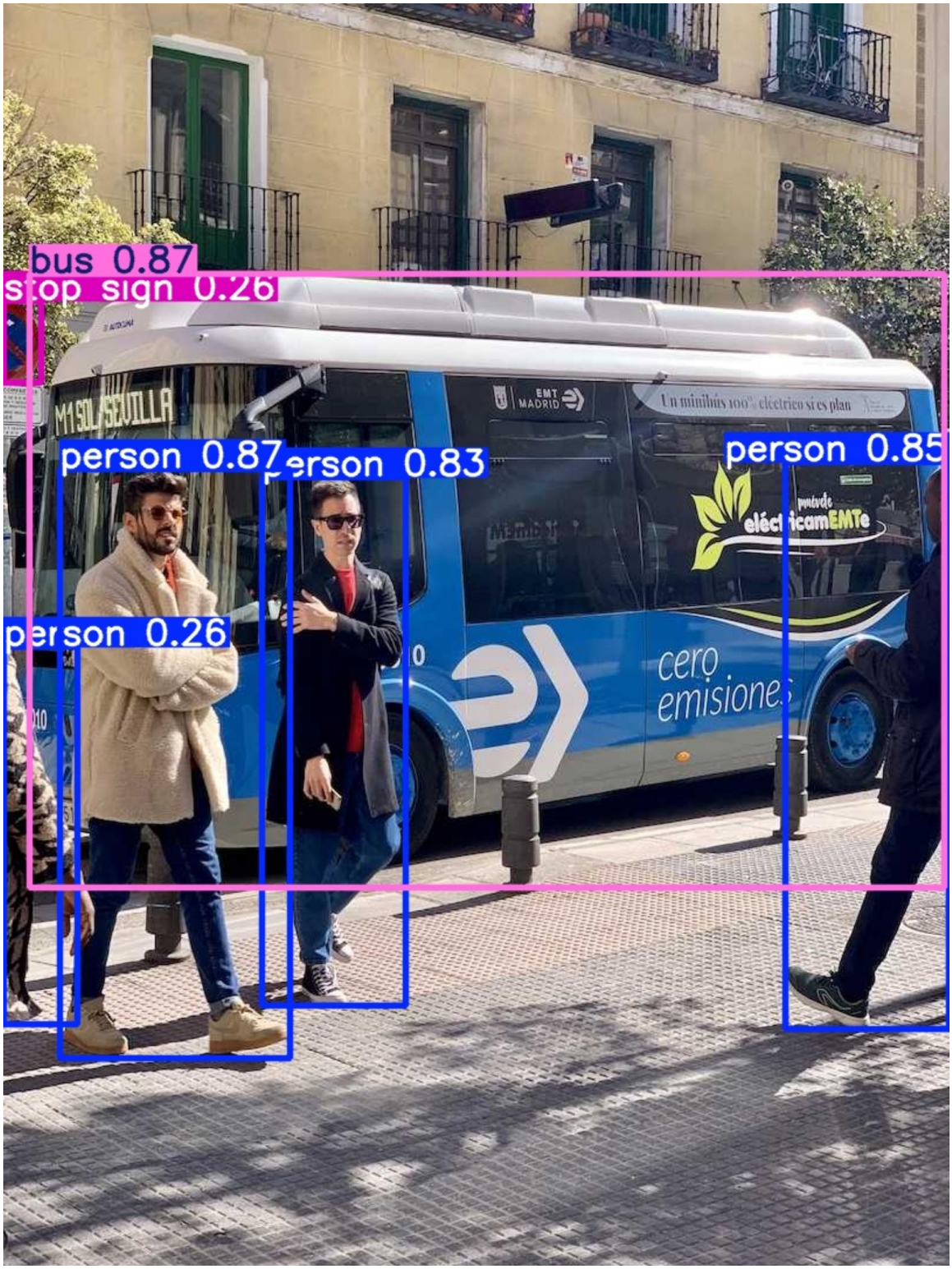

Figure 1: Object detection results showing bounding boxes around detected objects.

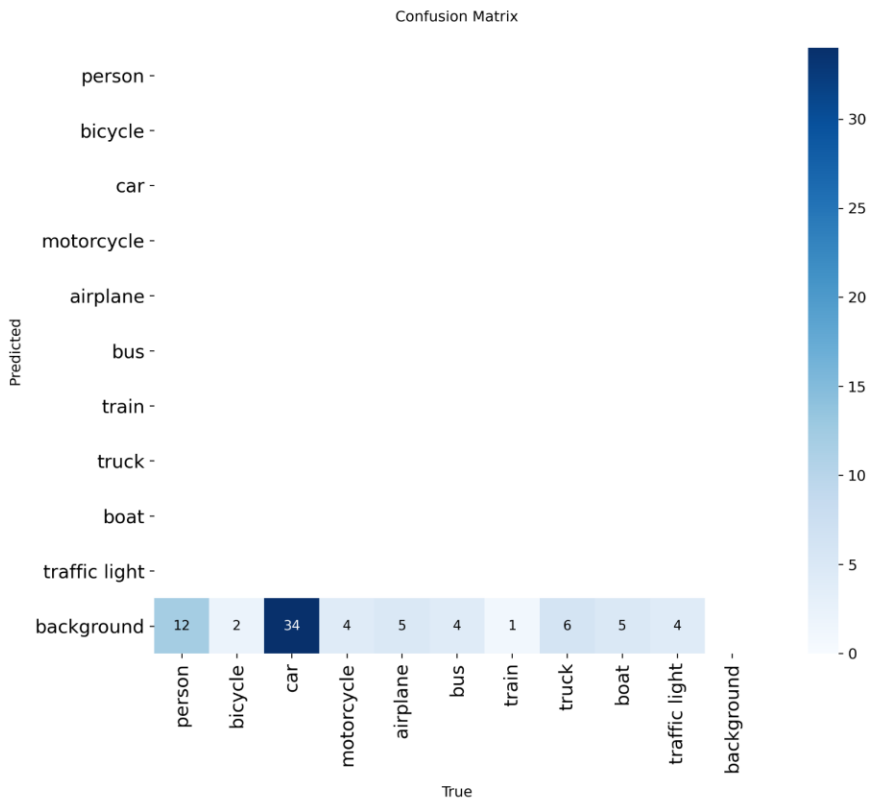

Figure 2: Confusion matrix representing classification performance.

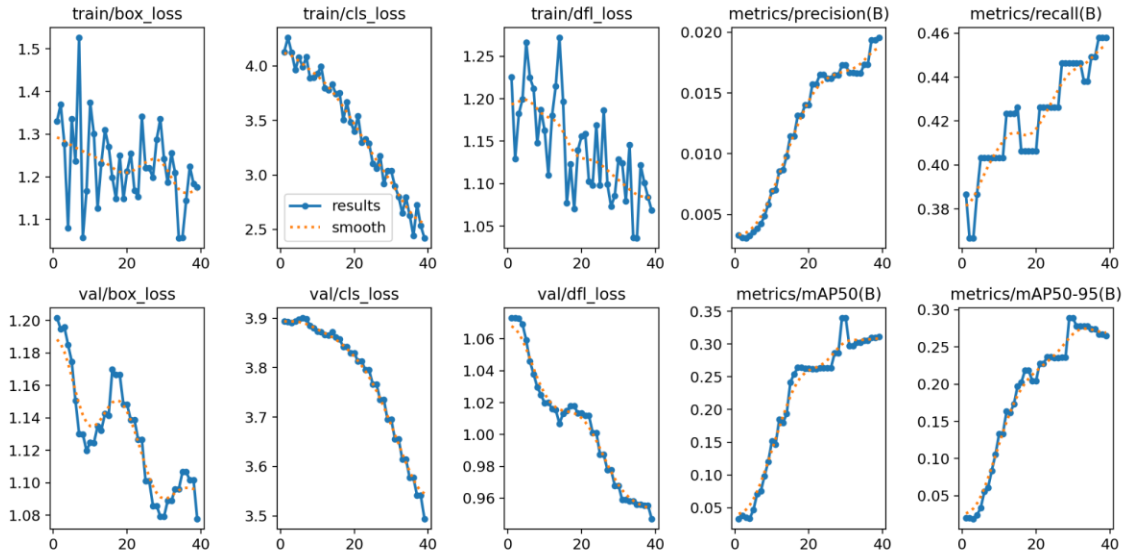

Figure 3: Training metrics including loss, precision, recall, and mAP.